Chapter 2: Respiratory diseases

Acute upper airway obstruction

Rhinitis and rhinopharyngitis (common cold)

Acute sinusitis

Acute pharyngitis

Diphtheria

Other upper respiratory tract infections

- Croup (laryngotracheitis and laryngotracheobronchitis)
- Epiglottitis
- Bacterial tracheitis

Otitis

- Acute otitis externa
- Acute otitis media (AOM)
- Chronic suppurative otitis media (CSOM)

Pertussis (whooping cough)

Bronchitis

- Acute bronchitis
- Chronic bronchitis

Bronchiolitis

Acute pneumonia

- Pneumonia in children under 5 years of age
- Pneumonia in children over 5 years and adults
- Persistent pneumonia

Staphylococcal pneumonia

Asthma

- Acute asthma (asthma attack)
- Chronic asthma

Pulmonary tuberculosis

Acute upper airway obstruction

Acute upper airway obstruction can be caused by foreign body aspiration, viral or bacterial infections (croup, epiglottitis, tracheitis), anaphylaxis, burns or trauma.
Initially stable and partial obstruction may worsen and develop into a life-threatening emergency, especially in young children.

Clinical features

Clinical signs of the severity of obstruction:

Obstruction	Signs	Danger signs
Complete	• Respiratory distress followed by cardiac arrest	Yes
Imminent complete	• Severe respiratory distress with cyanosis or $SpO_2 < 90\%$ • Agitation or lethargy • Tachycardia, capillary refill time > 3 seconds	
Severe	• Stridor (abnormal high pitched sound on inspiration) at rest • Severe respiratory distress: ▫ Severe intercostal and subcostal retractions ▫ Nasal flaring ▫ Substernal retractions (inward movement of the breastbone during inspiration) ▫ Severe tachypnoea	
Moderate	• Stridor with agitation • Moderate respiratory distress: ▫ Mild intercostal and subcostal retractions ▫ Moderate tachypnoea	No
Mild	• Cough, hoarse voice, no respiratory distress	

Management in all cases

- Examine children in the position in which they are the most comfortable.
- Evaluate the severity of the obstruction according to the table above.
- Monitor SpO_2, except in mild obstruction.
- Administer oxygen continuously:
  - to maintain the SpO_2 between 94 and 98% if it is ≤ 90% [a] or if the patient has cyanosis or respiratory distress;
  - if pulse oxymeter is not available: at least 5 litres/minute or to relieve the hypoxia and improve respiration.
- Hospitalize (except if obstruction is mild), in intensive care if danger signs.
- Monitor mental status, heart and respiratory rate, SpO_2 and severity of obstruction.
- Maintain adequate hydration by mouth if possible, by IV if patient unable to drink.

Management of foreign body aspiration

Acute airway obstruction (the foreign body either completely obstructs the pharynx or acts as a valve on the laryngeal inlet), no warning signs, most frequently in a child 6 months-5 years playing with a small object or eating. Conscience is initially maintained.

Perform maneuvers to relieve obstruction only if the patient cannot speak or cough or emit any sound:

- Children over 1 year and adults:
  Heimlich manoeuvre: stand behind the patient. Place a closed fist in the pit of the stomach, above the navel and below the ribs. Place the other hand over fist and press hard into the abdomen with a quick, upward thrust. Perform one to five abdominal thrusts in order to compress the lungs from the below and dislodge the foreign body.
- Children under 1 year:
  Place the infant face down across the forearm (resting the forearm on the leg) and support the infant's head with the hand. With the heel of the other hand, perform one to five slaps on the back, between shoulder plates.
  If unsuccessful, turn the infant on their back. Perform five forceful sternal compressions as in cardiopulmonary resuscitation: use 2 or 3 fingers in the center of the chest just below the nipples. Press down approximately one-third the depth of the chest (about 3 to 4 cm).

Repeat until the foreign body is expelled and the patient resumes spontaneous breathing (coughing, crying, talking). If the patient loses consciousness ventilate and perform cardiopulmonary rescucitation. Tracheostomy if unable to ventilate.

Differential diagnosis and management of airway obstructions of infectious origin

Infections	Symptoms	Appearance	Timing of symptoms
Viral croup	Stridor, cough and moderate respiratory difficulty	Prefers to sit	Progressive
Epiglottitis	Stridor, high fever and severe respiratory distress	Prefers to sit, drooling (cannot swallow their own saliva)	Rapid
Bacterial tracheitis	Stridor, fever, purulent secretions and severe respiratory distress	Prefers to lie flat	Progressive
Retropharyngeal or tonsillar abscess	Fever, sore throat and painful swallowing, earache, trismus and hot potato voice	Prefers to sit, drooling	Progressive

- Croup, epiglottitis, and tracheitis: see Other upper respiratory tract infections.
- Abscess: refer for surgical drainage.

Management of other causes

- Anaphylactic reaction (angioedema): see Anaphylactic shock (Chapter 1)
- Burns to the face or neck, smoke inhalation with airway oedema: see Burns (Chapter 10).

Footnotes

(a) If possible it is better to treat all patients with a $SpO_2 < 95\%$ with oxygen.

Rhinitis and rhinopharyngitis (common cold)

Rhinitis (inflammation of the nasal mucosa) and rhinopharyngitis (inflammation of the nasal and pharyngeal mucosa) are generally benign, self-limited and most often of viral origin. However, they may be an early sign of another infection (e.g. measles or influenza) or may be complicated by a bacterial infection (e.g. otitis media or sinusitis).

Clinical features

- Nasal discharge or obstruction, which may be accompanied by sore throat, fever, cough, lacrimation, and diarrhoea in infants. Purulent nasal discharge is not indicative of a secondary bacterial infection.
- In children under 5 years, routinely check the tympanic membranes to look for an associated otitis media.

Treatment

- Antibiotherapy is not recommended: it does not promote recovery nor prevent complications.
- Treatment is symptomatic:
  - Clear the nose with 0.9% sodium chloride[a] .
  - Fever, throat soreness: paracetamol PO for 2 to 3 days (Fever, Chapter 1).

Footnotes

(a) For a child: place him on his back, head turned to the side, and instil 0.9% sodium chloride into each nostril.

Acute sinusitis

Acute sinusitis is an inflammation of one or more of the sinus cavities, caused by an infection or allergy.
Most acute sinus infections are viral and resolve spontaneously in less than 10 days. Treatment is symptomatic.
Acute bacterial sinusitis may be a primary infection, a complication of viral sinusitis or of dental origin. The principal causative organisms are *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Moraxella catarrhalis*.
It is essential to distinguish between bacterial sinusitis and common rhinopharyngitis (see Rhinitis and rhinopharyngitis).
Antibiotic therapy is required in case of bacterial sinusitis only.
Without treatment, severe sinusitis in children may cause serious complications due to the spread of infection to the neighbouring bony structures, orbits or the meninges.

Clinical features

Sinusitis in adults

- Purulent unilateral or bilateral discharge, nasal obstruction
  and
- Facial unilateral or bilateral pain that increases when bending over; painful pressure in maxillary area or behind the forehead.
- Fever is usually mild or absent.

Sinusitis is likely if symptoms persist for longer than 10 to 14 days or worsen after 5 to 7 days or are severe (severe pain, high fever, deterioration of the general condition).

Sinusitis in children

- Same symptoms; in addition, irritability or lethargy or cough or vomiting may be present.
- In the event of severe infection: deterioration of the general condition, fever over 39 °C, periorbital or facial oedema.

Treatment

Symptomatic treatment

- Fever and pain (Chapter 1).
- Clear the nose with 0.9% sodium chloride[a] .

Antibiotherapy

- In adults:
  Antibiotherapy is indicated if the patient meets the criteria of duration or severity of symptoms. Oral amoxicillin is the first-line treatment.
  If the diagnosis is uncertain (moderate symptoms < 10 days) and the patient can be reexamined in the next few days, start with a symptomatic treatment, as for rhinopharyngitis or viral sinusitis.
- In children:
  Antibiotic therapy is indicated if the child has severe symptoms or mild symptoms associated with risk factors (e.g. immunosuppression, sickle cell disease, asthma).
  - Oral amoxicillin is the first-line treatment.
    amoxicillin PO for 7 to 10 days:

Children: 30 mg/kg 3 times daily (max. 3 g daily)
Adults: 1 g 3 times daily

- In the event of failure to respond within 48 hours of therapy:
  amoxicillin/clavulanic acid PO for 7 to 10 days. Use formulations in a ratio of 8:1 or 7:1 exclusively. The dose is expressed in amoxicillin:
  Children < 40 kg: 25 mg/kg 2 times daily
  Children ≥ 40 kg and adults:
  Ratio 8:1: 2000 mg daily (2 tablets of 500/62.5 mg 2 times daily)
  Ratio 7:1: 1750 mg daily (1 tablet of 875/125 mg 2 times daily)
- In penicillin-allergic patients:
  erythromycin PO for 7 to 10 days:
  Children: 30 to 50 mg/kg daily[b]
  Adults: 1 g 2 to 3 times daily

- In infants with ethmoiditis, see Periorbital and orbital cellulitis (Chapter 5).

Other treatments

- For sinusitis secondary to dental infection: dental extraction while under antibiotic treatment.
- In the event of ophthalmologic complications (ophthalmoplegia, mydriasis, reduced visual acuity, corneal anesthesia), refer for surgical drainage.

Footnotes

(a) For a child: place him on his back, head turned to the side, and instil 0.9% sodium chloride into each nostril.

(b) For dosage according to age or weight, see erythromycin in the guide Essential drugs, MSF.

Acute pharyngitis

Last updated: November 2020

Acute inflammation of the tonsils and pharynx. The majority of cases are of viral origin and do not require antibiotic treatment. Group A streptococcus (GAS) is the main bacterial cause, and mainly affects children aged 3 to 14 years. Acute rheumatic fever (ARF), a serious late complication of GAS pharyngitis, can be prevented with antibiotic treatment.
One of the main objectives of assessing acute pharyngitis is to identify patients requiring antibiotic treatment.

Clinical features

- Features common to all types of pharyngitis: throat pain, dysphagia (difficulty swallowing), inflammation of the tonsils and pharynx, tender anterior cervical lymph nodes, with or without fever.
- Specific features, depending on the cause:

  Common forms:
  - **Erythematous (red throat) or exudative (red throat and whitish exudate) pharyngitis**: this appearance is common to both viral and GAS pharyngitis. Centor criteria help assessment and decrease the empirical use of antibiotics in settings where rapid testing for GAS is not available. A Centor score of less than 2 rules out GAS infection[1][2] . Nevertheless, in patients with risk factors (immunosuppression, personal or family history of ARF) for poststreptococcal complications, or for local or general complications, do not use Centor score and prescribe empirical antibiotic treatment.

    Centor criteria

Criteria	Score
Temperature > 38 °C	1
Absence of cough	1
Tender anterior cervical lymph node(s)	1
Tonsillar swelling or exudate	1

    In patients over 14 years, the probability of GAS pharyngitis is low. Infectious mononucleosis (IM) due to the Epstein-Barr virus should be suspected in adolescents and young adults with extreme fatigue, generalized adenopathy and often splenomegaly.
    Erythematous or exudative pharyngitis may also be associated with gonococcal or primary HIV infection. In these cases, the diagnosis is mainly prompted by the patient's history.
  - **Pseudomembranous pharyngitis** (red tonsils/pharynx covered with an adherent greyish white false membrane): see Diphtheria, Chapter 2.
  - **Vesicular pharyngitis** (clusters of tiny blisters or ulcers on the tonsils): always viral (coxsackie virus or primary herpetic infection).
  - **Ulcero-necrotic pharyngitis**: hard and painless syphilitic chancre of the tonsil; tonsillar ulcer soft on palpation in a patient with poor oral hygiene and malodorous breath (Vincent tonsillitis).

  Other forms of pharyngitis:

- Spots on oral mucosa (Koplik's spots) accompanied by conjunctivitis and skin rash (see Measles, Chapter 8).
- "Strawberry" (red and bumpy) tongue accompanied by a skin rash: scarlet fever caused by GAS.

- Local complications:
  Peritonsillar, retropharyngeal or lateral pharyngeal abscess: fever, intense pain, dysphagia, hoarse voice, trismus (limitation of mouth opening), unilateral deviation of the uvula.

- General complications:
  - Complications due to the toxin: diphtheria (see Diphtheria, Chapter 2).
  - Poststreptococcal complications: ARF, acute glomerulonephritis.
  - Signs of serious illness in children*:* severe dehydration, severe difficulty swallowing, upper airway compromise, deterioration of general condition.

- Differential diagnosis: epiglottitis (see Epiglottitis, Chapter 2).

Treatment

- Symptomatic treatment (fever and pain): paracetamol or ibuprofen PO (Fever, Chapter 1).
- Centor score ≤ 1: viral pharyngitis, which typically resolves within a few days (or weeks, for IM): no antibiotic treatment.
- Centor score ≥ 2 or scarlet fever: antibiotic treatment for GAS infections[3] :
  - If single-use injection equipment is available, benzathine benzylpenicillin is the drug of choice as streptococcus A resistance to penicillin remains rare; it is the only antibiotic proven effective in reducing the incidence of rheumatic fever; and the treatment is administered as a single dose.
    benzathine benzylpenicillin IM
    Children under 30 kg (or under 10 years): 600 000 IU single dose
    Children 30 kg and over (or 10 years and over) and adults: 1.2 MIU single dose
  - Penicillin V is the oral reference treatment, but poor adherence is predictable due to the length of treatment.
    phenoxymethylpenicillin (penicillin V) PO for 10 days
    Children 1 to < 6 years: 250 mg 2 times daily
    Children 6 to < 12 years: 500 mg 2 times daily
    Children 12 years and over and adults: 1 g 2 times daily
    Children under 1 year: 125 mg 2 times daily
  - Amoxicillin is an alternative and the treatment has the advantage of being relatively short. However, it can cause adverse skin reactions in patients with undiagnosed IM and thus should be avoided when IM has not been excluded
    amoxicillin PO for 6 days
    Children: 25 mg/kg 2 times daily
    Adults: 1 g 2 times daily
  - Macrolides should be reserved for penicillin allergic patients as resistance to macrolides is frequent and their efficacy in the prevention of rheumatic fever has not been studied.
    azithromycin PO for 3 days
    Children: 20 mg/kg once daily (max. 500 mg daily)
    Adults: 500 mg once daily

- Gonococcal or syphilitic pharyngitis: as for genital gonorrhoea (Chapter 9) and syphilis (Chapter 9).
- Diphtherial pharyngitis: see Diphtheria (Chapter 2).
- Vincent tonsillitis: metronidazole or amoxicillin.
- Peritonsillar retropharyngeal or lateral pharyngeal abscess: refer for surgical drainage.

- If signs of serious illness or epiglottitis are present in children: hospitalise.

References

1. Fine AM, Nizet V, Mandl KD. Large-scale validation of the Centor and McIsaac scores to predict group A streptococcal pharyngitis. *Arch Intern Med.* 2012;172(11):847-852. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3627733/ [Accessed 20 October 2020]

2. National Institute for Health and Care Excellence. Sore throat (acute): antimicrobial prescribing. 2018. http://www.nice.org.uk/ng84 [Accessed 20 October 2020]

3. Group A Streptococcal Disease, Centers for Disease Control and Prevention. Atlanta (GA): CDC; 2020. https://www.cdc.gov/groupastrep/diseases-hcp/strep-throat.html [Accessed 20 October 2020]

Diphtheria

Last updated: October 2022

Diphtheria is a bacterial infection due to *Corynebacterium diphtheriae,* spread from person to person through inhalation of infected respiratory droplets of symptomatic or asymptomatic individuals, or direct contact with contaminated objects or diphtheria skin lesions[1][2][a] .
After infection, *C. diphtheriae* has an incubation period of 1 to 5 days (max. 10 days)[1] during which time it multiplies in the upper respiratory tract. The bacteria secretes a toxin which causes severe local as well as systemic effects. Death can occur from airway obstruction or as a result of systemic complications, including damage to the myocardium and nervous system, caused by the toxin.
Cases can remain infectious up to 8 weeks after initial infection[2] . Antibiotic treatment can reduce infectiousness to 6 days[3] .
Vaccination is the key to prevention and control of diphtheria. It protects individuals from severe disease (fewer and less severe symptoms) but does not prevent the spread of *C. diphtheriae*. Clinical disease does not confer protective immunity and vaccination is an integral part of case management.

Clinical features

- During clinical examination respect standard, contact, and droplet precautions (handwashing, gloves, gown, mask, etc.). Conduct a careful examination of the throat.
- Signs of respiratory diphtheria[a] :
  - pharyngitis, rhinopharyngitis, tonsillitis or laryngitis with tough, greyish, firmly adherent pseudo-membranes of the pharynx, nasopharynx, tonsils, or larynx;
  - dysphagia and cervical adenitis, at times progressing to massive swelling of the neck;
  - airway obstruction and possible suffocation when the infection extends to the nasal passages, larynx, trachea and bronchi;
  - fever is generally low-grade[2] .
- Generalised signs due to effects of the toxin:
  - cardiac dysfunction (tachycardia, arrhythmias), severe myocarditis with heart failure and possibly cardiogenic shock (see Shock, Chapter 1) 3 to 7 days or 2 to 3 weeks after onset of the disease;
  - neuropathies in 2 to 8 weeks after the onset of disease leading to nasal voice and difficulty with swallowing (paralysis of the soft palate), vision (ocular motor paralysis), breathing (paralysis of respiratory muscles) and ambulation (limb paralysis);
  - oliguria, anuria and acute renal failure.
- Differential diagnoses: Epiglottitis and Acute pharyngitis, Chapter 2, Stomatitis, Chapter 3.

Laboratory

- Diagnosis is confirmed by isolation of toxigenic *C. diphtheriae* by culture (and antibiotic susceptibility test) of swab specimens collected from the affected areas: throat (tonsils, pharyngeal mucosa, soft palate, exudate, ulcer, etc.), nasopharynx.
- The presence of the toxin is confirmed by PCR testing (detection of diphtheria toxin gene).

Treatment

- Isolation of patients; standard, droplet, and contact precautions for medical staff.

- **Diphtheria antitoxin** (**DAT**)[b] derived from horse serum:
  Administer DAT as soon as possible after disease onset. Do not wait for bacteriological confirmation[1] ; administer DAT under close monitoring in a hospital setting, according to the Besredka method to assess possibility of allergy. Any delay can diminish efficacy.
  ⚠ There is a risk of anaphylactic reaction, especially in patients with asthma. Close monitoring of the patient is essential, with immediate availability of equipment for manual ventilation (Ambu bag, face mask) and intubation, Ringer lactate and epinephrine (see Shock, Chapter 1).
  Besredka method: inject 0.1 ml SC and wait 15 minutes. If there is no allergic reaction (no erythema at the injection site or a flat erythema of less than 0.5 cm in diameter), inject a further 0.25 ml SC. If there is no reaction after 15 minutes, inject the rest of the product IM or IV depending on the volume to be administered.
  Doses are given as a function of the severity of illness, and the delay in treatment:

Clinical signs	Dose in units	Administration route
Laryngitis or pharyngitis or duration < 48 hours	20 to 40 000	IM or IV infusion in 250 ml of 0.9% sodium chloride in 2 to 4 hours for doses of more than 20 000 units.
Rhinopharyngitis	40 to 60 000	
Severe disease (respiratory distress, shock), cervical oedema or duration ≥ 48 hours	80 to 100 000	

- Antibiotic treatment (as soon as possible without waiting for bacteriological confirmation) for 14 days or according to length of treatment recommended by the national protocol:
  - if the patient can swallow:
    azithromycin PO (first-line)
    Children: 10 to 12 mg/kg once daily (max. 500 mg daily)
    Adults: 500 mg once daily
    or
    erythromycin PO
    Children under 40 kg: 10 to 15 mg/kg (max. 500 mg) 4 times daily
    Children 40 kg and over and adults: 500 mg 4 times daily
    or
    phenoxymethylpenicillin (**penicillin V**) PO
    Children under 40 kg: 10 to 15 mg/kg (max. 500 mg) 4 times daily
    Children 40 kg and over and adults: 500 mg 4 times daily
  - If the patient cannot swallow, start with one of the treatments below and change as soon as possible to oral route with one of the oral treatments above to complete 14 days of treatment:
    procaine benzylpenicillin IM
    Children under 25 kg: 50 000 IU/kg (= 50 mg/kg) once daily (max. 1.2 MIU = 1.2 g daily)
    Children 25 kg and over and adults: 1.2 MIU (= 1.2 g) once daily
    ⚠ Never administer procaine benzylpenicillin by IV injection or infusion.
    In penicillin-allergic patients, use erythromycin IV[c] .
- Intubation/tracheotomy if necessary (airway obstruction, respiratory failure, etc.).
- If the event of shock, see Shock, Chapter 1, for complementary treatment.
- Update every patient's vaccination status before hospital discharge (or during first visit, if receiving home-based care). If the patient has been administered DAT and can receive adequate home-based follow up after hospital discharge, wait 3 weeks after administration of DAT before vaccination.

Management of close contacts

Close contacts include household members living under the same roof and people who were directly exposed (less than one metre) to nasopharyngeal secretions of the patient on a regular basis (e.g. family or close friends, children in the same class, medical personnel) during the 5 days or nights prior to onset of symptoms of the case[4] .

- Collect nasal and pharyngeal swabs for culture before starting antibiotic prophylaxis; temperature and throat examination daily (10 days); exclusion from school or work until 48 hours after starting antibiotic prophylaxis. If symptoms of respiratory infection appear: treat immediately as a case of diphtheria.
- Antibiotic prophylaxis:
  benzathine benzylpenicillin IM
  Children under 30 kg: 600 000 IU single dose
  Children 30 kg and over and adults: 1.2 MIU single dose
  ⚠ Benzathine benzylpenicillin should never be administered by IV route.
  or azithromycin PO or erythromycin PO as above for 7 days.
- Check vaccination status:
  - if less than 3 injections received: complete vaccination schedule (see Prevention below);
  - if 3 injections received, with the last injection over one year ago: administer a booster dose immediately;
  - if 3 injections received, with the last injection less than one year ago: a booster dose is not immediately necessary.

Outbreak surveillance measures

- A suspected case of diphtheria is defined as a person with:
  - pharyngitis, rhinopharyngitis, tonsillitis and/or laryngitis

  AND
  - an adherent pseudo-membrane of the pharynx, nose, tonsils and/or larynx[1] .
- Isolate and treat suspect cases without delay. Collect swab samples before starting antibiotic treatment. Submit case notification to the public health authorities within 24 hours[1] .

Prevention

- Routine vaccination (EPI), for information: 3 doses of conjugate vaccine containing the higher potency (D) formulation of diphtheria toxoid as soon as possible as of 6 weeks of age and at 4 week intervals; D booster between 12 and 23 months, then between 4 and 7 years; booster with a vaccine containing a reduced dose (d) of diphtheria toxoid between 9 and 15 years[5] .
- Catch-up vaccination (individuals who have not received routine vaccination), for information:
  - children 1 to 6 years: 3 doses of conjugate vaccine containing the higher potency (D) formulation of diphtheria toxoid at least 4 weeks apart;
  - children 7 years and over and adults (including medical staff): 3 doses of conjugate vaccine containing a reduced dose (d) of diphtheria toxoid. Administer with a minimum interval of 4 weeks between first and second dose and an interval of at least 6 months between second and third dose (in the event of an outbreak this interval may be reduced to 4 weeks to achieve protection quicker).

  Administer 2 subsequent booster doses containing d at least 4 weeks apart[5] .

Footnotes

(a) This guide focuses on respiratory diphtheria and signs due to the toxin. It should be noted that cutaneous diphtheria is still a significant reservoir of *C. diphtheriae.*

(b) DAT reduces mortality and should be given to all diphtheria patients. However, as supply is very limited, it may be necessary to define criteria and reserve DAT for the treatment of patients who will benefit the most from it. DAT can be administered to pregnant women.

(c) **erythromycin** IV infusion (60 minutes)
Children: 12.5 mg/kg every 6 hours (max. 2 g daily); adults: 500 mg every 6 hours
Erythromycin powder (1 g) should be reconstituted in 20 ml of water for injection only. Then, dilute each dose of erythromycin in 10 ml/kg of 0.9% sodium chloride in children less than 20 kg and in a bag of 250 ml of 0.9% sodium chloride in children 20 kg and over and in adults. Do not dilute in glucose.

References

1. World Health Organization. Diphtheria. Vaccine-Preventable Diseases Surveillance Standards. 2018. https://www.who.int/immunization/monitoring_surveillance/burden/vpd/WHO_SurveillanceVaccinePreventable_04_Diphtheria_R2.pdf?ua=1 [Accessed 11 August 2020]

2. Tiwari TSP, Wharton M. Chapter 19: Diphtheria Toxoid. In: Plotkin SA, Orenstein WA, Offit PA, editors. Vaccines. 7th ed. Philadelphia, PA: Elsevier; 2018. p. 261–275.

3. Truelove SA, Keegan LT, Moss WJ, Chaisson LH, Macher E, Azman AS, Lessler J. Clinical and Epidemiological Aspects of Diphtheria: A Systematic Review and Pooled Analysis. Clin Infect Dis. 2020 Jun 24;71(1):89-97. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7312233/ [Accessed 24 November 2020]

4. Pan American Health Organization, World Health Organization. Diphtheria in the Americas - Summary of the situation 2018. Epidemiological Update Diphtheria. 16 April 2018. https://www.paho.org/hq/index.php?option=com_docman&view=download&category_slug=diphtheria-%098968&alias=44497-16-april-2018-diphtheria-epidemiological-update-497&Itemid=270&lang=en [Accessed 11 August 2020]

5. World Health Organization. Diphtheria vaccine: WHO position paper - August 2017. Weekly epidemiological record 2017; 92/(31):417–436. https://www.who.int/immunization/policy/position_papers/wer_31_diphtheria_updated_position_paper.pdf?ua=1 [Accessed 11 August 2020]

Other upper respiratory tract infections

- Croup (laryngotracheitis and laryngotracheobronchitis)
- Epiglottitis
- Bacterial tracheitis

Croup (laryngotracheitis and laryngotracheobronchitis)

Last updated: October 2024

Common viral respiratory infection with peak incidence amongst children between 6 months and 3 years.

Clinical features

- Typical barking cough, hoarse voice or cry.
- Inspiratory stridor (abnormal high pitched sound on inspiration):
  - Croup is considered mild if the stridor only occurs with agitation;
  - Croup is considered severe if there is stridor at rest, especially when it is accompanied by respiratory distress.
- Wheezing may also be present if the bronchi are involved.

Treatment

- In the absence of inspiratory stridor or intercostal, subcostal or sternal retractions, treat symptomatically: ensure adequate hydration, seek medical attention if symptoms worsen (e.g. respiratory difficulty, noisy breathing, inability to tolerate oral fluids).
- If stridor is only present with agitation (mild croup)[1] :
  - Assure adequate hydration.
  - Corticosteroids:
    - **dexamethasone** PO: 0.15 to 0.6 mg/kg (max. 16 mg)[a] single dose
    - or, if not available, **prednisolone** PO: 1 mg/kg single dose
  - Keep the child under observation at least 30 minutes after oral corticosteroid. Consider hospitalisation or longer observation (> 4 hours) if the child is less than 6 months old, or is dehydrated, or lives far from health facility.
- If danger signs are present (stridor at rest, respiratory distress, hypoxia) or the child is unable to drink, admit to hospital[1] :
  - Administer oxygen continuously if respiratory distress or SpO_2 < 92%: maintain SpO_2 between 94 and 98% (or if SpO_2 cannot be determined, at least 5 litres/minute).
  - Insert a peripheral IV line and provide IV hydration.
  - **Epinephrine** (**adrenaline**) via nebulizer[2]b : 0.5 mg/kg (max. 5 mg) to be repeated every 20 minutes if danger signs persist (see table below).
    Monitor heart rate during nebulization (if heart rate greater than 200, stop the nebulization).

Weight	6 kg	7 kg	8 kg	9 kg	10-17 kg
Dose in mg	3 mg	3.5 mg	4 mg	4.5 mg	5 mg
Dose in ml (1 mg/ml, 1 ml ampoule)	3 ml	3.5 ml	4 ml	4.5 ml	5 ml
NaCl 0.9% (a)	1 ml	1 ml	–	–	–

(a) Add sufficient NaCl 0.9% to obtain a total volume of 4 to 4.5 ml in the nebulizing chamber.

⚠ Epinephrine is intended exclusively for nebulized administration and should not be given IV or IM in croup.

- Corticosteroids:
  - **dexamethasone**[c] PO (or IM or IV if the child is vomiting): 0.6 mg/kg (max. 16 mg) single dose (see table below)
  - or, if not available, **prednisolone** PO: 1 mg/kg single dose

Weight	6-8 kg	9-11 kg	12-14 kg	15-17 kg
Dose in mg	4 mg	6 mg	8 mg	10 mg
Dose in 2 mg tablet	2 tab	3 tab	4 tab	5 tab
Dose in ml (4 mg/ml, 1 ml ampoule)	1 ml	1.5 ml	2 ml	2.5 ml

- Suspect bacterial tracheitis in a critically ill appearing child[d] with croup who does not improve with the above treatment.
- If the patient has a complete airway obstruction, intubation if possible or emergency tracheotomy.

Footnotes

(a) If children can easily return to hospital in case of deterioration or return of symptoms, administer 0.15 mg/kg of dexamethasone. Otherwise, the dose of 0.6 mg/kg should be used.

(b) Although not licensed for this indication, epinephrine 1:1000 (1 mg/ml) should be used for nebulisation.

(c) Administer orally if possible in order to avoid causing agitation in the child as this may worsen symptoms.

(d) Critically ill appearing child: weak grunting or crying, drowsiness, difficult to arouse, does not smile, disconjugate or anxious gaze, pallor or cyanosis, general hypotonia.

References

1. Pocket book of primary health care for children and adolescents: guidelines for health promotion, disease prevention and management from the newborn period to adolescence. Copenhagen: WHO Regional Office for Europe; 2022. Licence: CC BY-NCSA 3.0 IGO. https://www.who.int/europe/publications/i/item/9789289057622

2. Bjornson C, Russell K, Vandermeer B, Klassen TP, Johnson DW. Nebulized epinephrine for croup in children. Cochrane Acute Respiratory Infections Group, ed. Cochrane Database Syst Rev. Published online October 10, 2013. https://doi.org/10.1002/14651858.CD006619.pub3

Epiglottitis

Bacterial infection of the epiglottis in young children caused by *Haemophilus influenzae* (Hib), it is rare when Hib vaccine coverage is high. It can be caused by other bacteria and occur in adults.

Clinical features

- Rapid (less than 12-24 hours) onset of high fever.
- Typical “tripod or sniffing” position, preferring to sit, leaning forward with an open mouth, anxious appearing.
- Difficulty swallowing, drooling, and respiratory distress.
- Stridor may be present (as opposed to croup, hoarse voice and cough are usually absent).
- Critically ill appearing[a] .

Allow the child to sit in a comfortable position or on the parent’s lap. Do not force them to lie down (may precipitate airway obstruction). Avoid any examination that will upset the child including examination of the mouth and throat.

Treatment

- In case of imminent airway obstruction, emergency intubation or tracheotomy is indicated. The intubation is technically difficult and should be performed under anaesthesia by a physician familiar with the procedure. Be prepared to perform a tracheotomy if intubation is unsuccessful.
- In all cases:
  - Insert a peripheral IV line and provide IV hydration.
  - Antibiotherapy:
    ceftriaxone slow IV (3 minutes) or IV infusion (30 minutes)[b] . Avoid IM route (may agitate the child and precipitate a respiratory arrest).
    Children: 50 mg/kg once daily
    Adults: 1 g once daily
    The IV treatment is administered for at least 5 days then, if the clinical condition has improved[c] and oral treatment can be tolerated, change to:
    amoxicillin/clavulanic acid (**co-amoxiclav**) PO to complete a total of 7 to 10 days of treatment. Use formulations in a ratio of 8:1 or 7:1 exclusively. The dose is expressed in amoxicillin:
    Children < 40 kg: 50 mg/kg 2 times daily
    Children ≥ 40 kg and adult:
    Ratio 8:1: 3000 mg daily (2 tablets of 500/62.5 mg 3 times daily)
    Ratio 7:1: 2625 mg daily (1 tablet of 875/125 mg 3 times daily)

Footnotes

(a) Critically ill appearing child: weak grunting or crying, drowsiness, difficult to arrouse, does not smile, unconjugate or anxious gaze, pallor or cyanosis, general hypotonia.

(b) For administration by IV route, ceftriaxone powder should to be reconstituted in water for injection only. For administration by IV infusion, dilute each dose of ceftriaxone in 5 ml/kg of 0.9% sodium chloride or 5% glucose in children less than 20 kg and in a bag of 100 ml of 0.9% sodium chloride or 5% glucose in children over 20 kg and in adults.

(c) Improvement criteria include: fever reduction, diminished respiratory distress, improved SpO_2, improved appetite and/or activity.

Bacterial tracheitis

Bacterial infection of the trachea in children, occurring as a complication of a previous viral infection (croup, influenza, measles, etc.).

Clinical features

- Fever in a critically ill appearing child[a] .
- Stridor, cough and respiratory distress.
- Copious purulent secretions.
- As opposed to epiglottitis the onset of symptoms is gradual and the child prefers to lie flat.
- In severe cases there is a risk of complete airway obstruction, especially in very young children.

Treatment

- Suction purulent secretions.
- Insert a peripheral IV line and provide IV hydration.
- Antibiotherapy:
  ceftriaxone slow IV[b] (3 minutes) or IV infusion (30 minutes). Do not administer by IM route (may agitate the child and precipitate a respiratory arrest).
  Children: 50 mg/kg once daily
  Adults: 1 g once daily
  +
  cloxacillin IV infusion (60 minutes)
  Children less than 12 years: 25 to 50 mg/kg every 6 hours
  Children 12 years and over and adults: 2 g every 6 hours
  The IV treatment is administered for at least 5 days then, if the clinical condition has improved[c] and oral treatment can be tolerated, change to :
  amoxicillin/clavulanic acid (**co-amoxiclav**) PO to complete 7 to 10 days of treatment, as in epiglottitis.
- If the event of complete airway obstruction, intubation if possible or emergency tracheotomy.

Footnotes

(a) Critically ill appearing child: weak grunting or crying, drowsiness, difficult to arrouse, does not smile, unconjugate or anxious gaze, pallor or cyanosis, general hypotonia.

(b) For administration by IV route, ceftriaxone powder should to be reconstituted in water for injection only. For administration by IV infusion, dilute each dose of ceftriaxone in 5 ml/kg of 0.9% sodium chloride or 5% glucose in children less than 20 kg and in a bag of 100 ml of 0.9% sodium chloride or 5% glucose in children over 20 kg and in adults.

(c) Improvement criteria include: fever reduction, diminished respiratory distress, improved SpO_2, improved appetite and/or activity.

Otitis

- Acute otitis externa
- Acute otitis media (AOM)
- Chronic suppurative otitis media (CSOM)

Acute otitis externa

Diffuse inflammation of the external ear canal, due to bacterial or fungal infection. Common precipitants of otitis externa are maceration, trauma of the ear canal or presence of a foreign body or dermatologic diseases (such as eczema, psoriasis).

Clinical features

- Ear canal pruritus or ear pain, often severe and exacerbated by motion of the pinna; feeling of fullness in the ear; clear or purulent ear discharge or no discharge
- Otoscopy (remove skin debris and secretions from the auditory canal by gentle dry mopping (use a dry cotton bud or a small piece of dry cotton wool):
  - diffuse erythema and edema, or infected eczema, of the ear canal
  - look for a foreign body
  - if visible, the tympanic membrane is normal (swelling and pain very often prevent adequate visualization of the tympanic membrane)

Treatment

- Remove a foreign body, if present.
- Treatment of pain: paracetamol PO (Chapter 1, Pain).
- Local treatment:
  - Remove secretions from the auditory canal by gentle dry mopping (use a dry cotton bud or a small piece of dry cotton wool). Consider ear irrigation (0.9% sodium chloride, using a syringe) only if the tympanic membrane can be fully visualised and is intact (no perforation). Otherwise, ear irrigation is contra-indicated.
  - Apply **ciprofloxacin** ear drops in the affected ear(s) for 7 days:
    Children ≥ 1 year: 3 drops 2 times daily
    Adults: 4 drops 2 times daily

Acute otitis media (AOM)

Acute inflammation of the middle ear, due to viral or bacterial infection, very common in children under 3 years, but uncommon in adults.
The principal causative organisms of bacterial otitis media are *Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis* and in older children, *Streptococcus pyogenes*.

Clinical features

- Rapid onset of ear pain (in infants: crying, irritability, sleeplessness, reluctance to nurse) and ear discharge (otorrhoea) or fever.
- Other signs such as rhinorrhoea, cough, diarrhoea or vomiting are frequently associated, and may confuse the diagnosis, hence the necessity of examining the tympanic membranes.
- Otoscopy: bright red tympanic membrane (or yellowish if rupture is imminent) and presence of pus, either externalised (drainage in ear canal if the tympanic membrane is ruptured) or internalised (opaque or bulging tympanic membrane). The combination of these signs with ear pain or fever confirms the diagnosis of AOM.
  Note:
  The following otoscopic findings are not sufficient to make the diagnosis of AOM:
  - A red tympanic membrane alone, with no evidence of bulging or perforation, is suggestive of viral otitis in a context of upper respiratory tract infection, or may be due to prolonged crying in children or high fever.
  - The presence of air bubbles or fluid behind an intact tympanic membrane, in the absence of signs and symptoms of acute infection, is suggestive of otitis media with effusion (OME).
- Complications, particularly in high-risk children (malnutrition, immunodeficiency, ear malformation) include chronic suppurative otitis media, and rarely, mastoiditis, brain abscess or meningitis.

Treatment

- In all cases:
  - Treatment of fever and pain: paracetamol PO (Chapter 1).
  - Ear irrigation is contra-indicated if the tympanic membrane is ruptured, or when the tympanic membrane cannot be fully visualised. Ear drops are not indicated.
- Indications for antibiotic therapy:
  - Antibiotics are prescribed in children less than 2 years, children whose assessment suggests severe infection (vomiting, fever > 39 °C, severe pain) and children at risk of unfavourable outcome (malnutrition, immunodeficiency, ear malformation).
  - For other children:
    - If the child can be re-examined within 48 to 72 hours: it is preferable to delay antibiotic prescription. Spontaneous resolution is probable and a short symptomatic treatment of fever and pain may be sufficient. Antibiotics are prescribed if there is no improvement or worsening of symptoms after 48 to 72 hours.
    - If the child cannot be re-examined: antibiotics are prescribed.
  - For children treated with antibiotics: advise the mother to bring the child back if fever and pain persist after 48 hours.
- Choice of antibiotherapy:
  - Amoxicillin is the first-line treatment:
    amoxicillin PO for 5 days
    Children: 30 mg/kg 3 times daily (max. 3 g daily)
    Adults: 1 g 3 times daily

- Amoxicillin/clavulanic acid is used as second-line treatment, in the case of treatment failure. Treatment failure is defined as persistence of fever and/or ear pain after 48 hours of antibiotic treatment.
  amoxicillin/clavulanic acid (**co-amoxiclav**) PO for 5 days
  Use formulations in a ratio of 8:1 or 7:1. The dose is expressed in amoxicillin:
  Children < 40 kg: 25 mg/kg 2 times daily
  Children ≥ 40 kg and adult:
  Ratio 8:1: 2000 mg daily (2 tablets of 500/62.5 mg 2 times daily)
  Ratio 7:1: 1750 mg daily (1 tablet of 875/125 mg 2 times daily)
  Persistence of a ear drainage alone, without fever and pain, in a child who has otherwise improved (reduction in systemic symptoms and local inflammation) does not warrant a change in antibiotic therapy. Clean ear canal by gentle dry mopping until no more drainage is obtained.
- Macrolides should be reserved for very rare penicillin-allergic patients, as treatment failure (resistance to macrolides) is frequent.
  azithromycin PO
  Children over 6 months: 10 mg/kg once daily for 3 days

Chronic suppurative otitis media (CSOM)

Chronic bacterial infection of the middle ear with persistent purulent discharge through a perforated tympanic membrane.
The principal causative organisms are *Pseudomonas aeruginosa, Proteus* sp, staphylococcus, other Gram negative and anaerobic bacteria.

Clinical features

- Purulent discharge for more than 2 weeks, often associated with hearing loss or even deafness; absence of pain and fever
- Otoscopy: perforation of the tympanic membrane and purulent exudate
- Complications:
  - Consider a superinfection (AOM) in the case of new onset of fever with ear pain, and treat accordingly.
  - Consider mastoiditis in the case of new onset of high fever, severe ear pain and/or tender swelling behind the ear, in a patient who appears significantly unwell.
  - Consider brain abscess or meningitis in the case of impaired consciousness, neck stiffness and focal neurological signs (e.g. facial nerve paralysis).

Treatment

- Remove secretions from the auditory canal by gentle dry mopping (use a dry cotton bud or a small piece of dry cotton wool).
- Apply **ciprofloxacin** ear drops until no more drainage is obtained (approximately 2 weeks, max. 4 weeks):
  Children 1 year and over: 3 drops 2 times daily
  Adults: 4 drops 2 times daily
- Complications:
  - Chronic mastoiditis is a medical emergency that requires prompt hospitalisation, prolonged antibiotherapy that covers the causative organisms of CSOM (**ceftriaxone** IM for 10 days + **ciprofloxacin** PO for 14 days), atraumatic cleaning of the ear canal; surgical treatment may be required. Before transfer to hospital, if the patient needs to be transferred, administer the first dose of antibiotics.
  - Meningitis (Chapter 7).

Pertussis (whooping cough)

Last updated: December 2024

Pertussis is a highly contagious infection of the respiratory tract caused by the bacterium *Bordetella pertussis*. It is transmitted via airborne droplets spread by infected people (by coughing and/or sneezing), particularly within 3 weeks of the onset of cough[1] . Despite the development of effective vaccines, it remains an important cause of morbidity and mortality, particularly in infants.

This disease is endemic worldwide and outbreaks are common. It most commonly affects children < 5 years, but can affect any age group. Reported incidence in adolescents and adults has been increasing in some regions[2] .

Pertussis is a notifiable disease in many countries, and cases should be reported to local or national public health authorities. Pertussis immunization is part of all vaccination schedules and is included in the Expanded Programme on Immunization (EPI).

Clinical features

After an incubation period of 7 to 10 days (up to 21 days), the illness evolves in 3 stages[3] :

- Catarrhal stage (1 to 2 weeks, but may be shorter in infants < 6 months):
  - Runny nose, mild cough, no or low-grade fever.
  - During this stage, the disease is indistinguishable from other non-specific respiratory infections.
- Paroxysmal stage (1 to 6 weeks):
  - Cough of increasing severity, occuring in characteristic bouts (paroxysms) of a series of coughs during one exhalation, followed by a laboured inspiration causing a distinctive gasping sound (whoop), and/or post-tussive vomiting.
  - No or low-grade fever.
  - Apnoea and cyanosis (in infants).
  - Prolonged cough, sometimes without paroxysms or whoop (particularly in older children and adults).

  Complications may include secondary bacterial pneumonia (new-onset fever can be an indicator), dehydration and malnutrition triggered by feeding difficulty due to cough and vomiting, seizures, encephalopathy, sudden death, intracranial bleeding, petechiae, rib fracture, hernia, rectal prolapse.
- Convalescent stage (weeks to months):
  - Paroxysms of cough gradually decrease in frequency and severity.

The disease is most severe, with a high risk of death, in infants. Adolescents and adults commonly have milder symptoms, but older adults, immunocompromised persons, and people with underlying respiratory conditions are at risk of severe disease and hospitalisation[4] .

Pertussis is most contagious and antibiotic treatment is most beneficial during the catarrhal stage of the illness. Since this is also the period during which clinical symptoms and signs are non-specific, it is necessary to maintain a high index of suspicion with clues from the context. These include patients with any compatible symptoms and close contact with a suspected case or in a pertussis outbreak setting.

Management

Suspected cases

Antibiotic treatment

Antibiotic treatment is indicated for the following patients[5] and should be started as soon as pertussis is suspected:

- All patients 1 year of age and older, within 3 weeks of the onset of cough.
- Infants and pregnant women, within 6 weeks of the onset of cough.

First line	
azithromycin PO for 5 days	• Children < 6 months: 10 mg/kg once daily • Children ≥ 6 months: 10 mg/kg (max. 500 mg) on D1 then 5 mg/kg (max. 250 mg) once daily from D2 to D5 • Adults: 500 mg on D1 then 250 mg once daily from D2 to D5
Alternatives	
erythromycin PO[(a)] for 7 days	• Children: 15 mg/kg 3 times daily • Adults: 1 g 3 times daily
co-trimoxazole PO[(b)] for 14 days	• Children ≥ 6 weeks: 20 mg SMX/4 mg TMP/kg (max. 800 mg SMX/160 mg TMP) 2 times daily • Adults: 800 mg SMX/160 mg TMP 2 times daily

(a) Erythromycin is an alternative, but azithromycin is better tolerated and simpler to administer (shorter treatment duration, fewer daily doses).

(b) Use co-trimoxazole only if macrolides are contra-indicated, not tolerated, or not available.
Avoid:
- during the first trimester of pregnancy (risk of congenital malformations),
- after 36 weeks of pregnancy and in women breastfeeding neonates (0 to 4 weeks), or breastfeeding infants that are premature, low birth weight, jaundiced, or ill (risk of haemolysis and jaundice in the child).
Advise patients to seek immediate medical attention if there are signs of anaemia or jaundice, especially in regions with high prevalence of G6PD deficiency.

Admit the following patients to hospital

- Patients with severe illness (e.g. respiratory distress, apnoea, cyanosis, pneumonia, seizures or impaired consciousness)
- Infants up to 3 months of age (with monitoring 24 hours per day due to the risk of apnoea)
- Patients with difficulty in feeding or drinking

Infection prevention and control measures

Patients with pertussis are considered infectious until they have completed 5 days of appropriate antibiotics, or if not treated, until 21 days after the start of the paroxysmal cough[1] . During this infectious period, these measures should be followed:

- In hospital: single room if possible, or grouping pertussis cases together away from other patients (cohorting); standard precautions and droplet precautions (including, for the patient, cough etiquette and wearing a surgical mask if outside the room)[6]
- All patients: avoid contact with young children and pregnant women; avoid congregate settings including childcare, school, and work

Supportive care

Hydration and nutrition: if child is breastfeeding, continue; ensure adequate fluid and calorie intake, give frequent small feedings including after coughing bouts and post-tussive vomiting; some patients may require nasogastric tube feeding or IV maintenance fluids; be aware that NG tube placement might provoke paroxysmal coughing, so it should be inserted by experienced medical staff with minimal manipulation. If weight loss occurred during illness, consider food supplements for several weeks during convalescent stage when child is able to eat comfortably.

Respiratory measures

- Place the patient in a semi-reclining position (± 30°).
- In case of apnoea, stimulate the patient, but be prepared to ventilate if necessary (keep bag and mask accessible).
- Administer oxygen if $SpO_2 < 92\%$, for severe respiratory distress, or for recurrent apnoea. Perform gentle oropharyngeal suctioning if required but avoid deep suctioning (may provoke paroxysmal coughing).

For children treated as outpatients, teach the parents about signs that require immediate medical attention (e.g. respiratory difficulty, apnoea, cyanosis, increasing fever, seizures or impaired consciousness, dehydration, feeding difficulty).

Contacts

- Post-exposure prophylaxis (same antibiotic treatment as for suspect cases) is recommended, regardless of vaccination status, for:
  - Asymptomatic close contacts (having face-to-face exposure or direct contact with oral, nasal, or other respiratory secretions[7]) of a person with suspected pertussis within 3 weeks of onset of paroxysmal cough
  - Asymptomatic exposed people at high risk of complications or who will come into contact with people at high risk[4][8]:
    - pregnant women in the third trimester (risk to neonate),
    - infants under 1 year,
    - people with immunodepression, moderate to severe asthma, and consider for people with other underlying respiratory conditions and adults ≥ 65 years.
- Isolation of asymptomatic contacts is not necessary.
- Symptomatic contacts should be treated as suspected pertussis.
- Assess whether the person is up-to-date with pertussis vaccination according to national protocol (see Prevention).

Prevention

- Pertussis vaccination provides substantial immunity that wanes over time[1].
- In all cases (suspected cases and contacts) who are not up-to-date with pertussis vaccination, begin or refer for vaccination.
- Routine vaccination with a 3-dose primary series of polyvalent vaccine containing pertussis antigen(s) from the age of 6 weeks or according to national protocol. A booster dose is recommended, preferably during the second year of life[1]. Based on local epidemiology, further booster doses may be required later in life to reinforce immunity and reduce the risk of developing pertussis.
- If the pertussis primary vaccination series has been interrupted, it should be completed rather than restarted from the beginning.
- In general, vaccinations are deferred for patients with moderate to severe illness, but the vaccine should be administered as soon as the patient's condition has improved.
- Vaccination will probably not prevent disease in a person who is already infected by *B. pertussis*. It is not a substitute for post-exposure prophylaxis[7][9].

References

1. Pertussis vaccines: WHO position paper – August 2015. *Wkly Epidemiol Rec*. 2015;(35):433-458. https://iris.who.int/bitstream/handle/10665/242413/WER9035_433-458.PDF [Accessed May 17, 2024].

2. Güris D, Strebel PM, Bardenheier B, et al. Changing Epidemiology of Pertussis in the United States: Increasing Reported Incidence Among Adolescents and Adults, 1990-1996. *Clin Infect Dis*. 1999;28(6):1230-1237. https://doi.org/10.1086/514776 [Accessed September 10, 2024].

3. Daniels HL, Sabella C. *Bordetella pertussis* (Pertussis). *Pediatr Rev*. 2018;39(5):247-257. https://doi.org/10.1542/pir.2017-0229 [Accessed September 11, 2024].

4. Mbayei SA, Faulkner A, Miner C, et al. Severe Pertussis Infections in the United States, 2011–2015. *Clin Infect Dis*. 2019;69(2):218-226. https://doi.org/10.1093/cid/ciy889 [Accessed September 13, 2024]

5. Centers for Disease Control and Prevention. For Clinicians: Pertussis Treatment. Published online August 4, 2022. https://www.cdc.gov/pertussis/clinical/treatment.html [Accessed April 28, 2024].

6. Siegel J, Rhinehart E, Jackson M, Chiarello L, and the Healthcare Infection Control Practices Advisory Committee. 2007 Guideline for Isolation Precautions: Preventing Transmission of Infectious Agents in Healthcare Settings. Published online July 2023. https://www.cdc.gov/infection-control/media/pdfs/Guideline-Isolation-H.pdf [Accessed April 28,2024].

7. World Health Organization. Pertussis: Vaccine-Preventable Diseases Surveillance Standards. Published online September 5, 2018. https://www.who.int/publications/m/item/vaccine-preventable-diseases-surveillance-standards-pertussis [Accessed May 19, 2024].

8. Centers for Disease Control and Prevention. Pertussis and Postexposure Antimicrobial Prophylaxis (PEP). Published online August 4, 2022. https://www.cdc.gov/pertussis/php/postexposure-prophylaxis/index.html [Accessed April 29, 2024].

9. Blain A, Skoff T, Cassiday P, Tondella ML, Acosta A. Pertussis. In: *Manual for the Surveillance of Vaccine-Preventable Diseases*. Centers for Disease Control and Prevention; 2020:Chapter 10. https://www.cdc.gov/surv-manual/php/table-of-contents/chapter-10-pertussis.html [Accessed May 19, 2024].

Bronchitis

- Acute bronchitis
- Chronic bronchitis

Acute bronchitis

An acute inflammation of the bronchial mucosa, most commonly of viral origin. In older children it can be caused by *Mycoplasma pneumoniae*. In children over 2 years of age with repetitive acute bronchitis or 'wheezing' bronchitis, consider asthma (see Asthma). In children under 2 years of age, consider bronchiolitis (see Bronchiolitis).

Clinical features

Often begins with a rhinopharyngitis that descends progressively: pharyngitis, laryngitis, tracheitis.

- Heavy cough, dry at the beginning then becoming productive
- Low-grade fever
- No tachypnoea, no dyspnoea
- On pulmonary auscultation: bronchial wheezing

Treatment

- Fever: paracetamol PO (Chapter 1).
- Keep the patient hydrated, humidify air (with a bowl of water or a wet towel).
- Children: nasal irrigation with 0.9% sodium chloride or Ringer lactate, 4 to 6 times daily to clear the airway.
- Antibiotherapy is not useful for patients in good overall condition with rhinopharyngitis or influenza.
- Antibiotherapy is indicated only if:
  - the patient is in poor general condition: malnutrition, measles, rickets, severe anaemia, cardiac disease, elderly patient etc.
  - if the patient has dyspnoea, fever greater than 38.5 °C and purulent expectorations: a secondary infection with *Haemophilus influenzae* or with pneumococcus is probable.
    amoxicillin PO
    Children: 30 mg/kg 3 times daily (max. 3 g daily) for 5 days
    Adults: 1 g 3 times daily for 5 days

Chronic bronchitis

A chronic inflammation of the bronchial mucosa due to irritation (tobacco, pollution), allergy (asthma) or infection (repetitive acute bronchitis). It may develop into chronic obstructive pulmonary disease.

Clinical features

- Productive cough for 3 consecutive months per year for 2 successive years.
- No dyspnoea at onset. Dyspnoea develops after several years, first on exertion, then becoming persistent.
- On pulmonary auscultation: bronchial wheeze (always exclude tuberculosis).

A patient with an acute exacerbation of chronic bronchitis presents with:

- Onset or increase of dyspnoea.
- Increased volume of sputum.
- Purulent sputum.

Treatment

- Antibiotic treatment is not useful in treating simple chronic bronchitis.
- Antibiotic treatment may be useful, for patients in a poor general condition only, for acute exacerbations of chronic bronchitis (see Acute bronchitis).
- Discourage smoking and other irritating factors.

Bronchiolitis

Last updated: October 2023

Bronchiolitis is an epidemic and seasonal viral infection of the lower respiratory tract in children less than 2 years of age, characterised by bronchiolar obstruction.
Respiratory syncytial virus (RSV) is responsible for 70% of cases of bronchiolitis. Transmission of RSV is direct, through inhalation of droplets (coughing, sneezing), and indirect, through contact with hands or materials contaminated by infected secretions.
In the majority of cases, bronchiolitis is benign, resolves spontaneously (relapses are possible), and can be treated on an outpatient basis.
Severe cases may occur, which put the child at risk due to exhaustion or secondary bacterial infection. Hospitalisation is necessary when signs/criteria of severity are present (10 to 20% of cases).

Clinical features

- Tachypnoea, dyspnoea, wheezing, cough; profuse, frothy, obstructive secretions.
- On auscultation: prolonged expiration with diffuse, bilateral wheezes; sometimes diffuse fine, end-inspiratory crackles.

Rhinopharyngitis, with dry cough, precedes these features by 24 to 72 hours; fever is absent or moderate.

- Signs of severity:
  - Significant deterioration in general condition, toxic appearance (pallor, greyish colouration)
  - Apnoea, cyanosis (check lips, buccal mucosa, fingernails)
  - Respiratory distress (nasal flaring, sternal and chest wall indrawing)
  - Anxiety and agitation (hypoxia), altered level of consciousness
  - Respiratory rate > 60/minute
  - Decreased signs of respiratory distress (exhaustion) and decline of respiratory rate (< 30/minute below the age of 1 year and < 20/minute below the age of 3 years). Exercise caution in interpreting these signs as indicators of clinical improvement.
  - SpO_2 persistently < 92%
  - Sweats, tachycardia at rest and in the absence of fever
  - Silence on auscultation (severe bronchospasm)
  - Difficulty drinking or sucking (reduced tolerance for exertion)

Treatment

Treatment is symptomatic. Obstructive signs and symptoms last for about 10 days; cough may persist for 2 weeks or longer.

Hospitalise children with one of the following criteria:
- Presence of any sign of severity
- Pre-existing pathology (cardiac or pulmonary disease, malnutrition, HIV infection, etc.)

Consider hospitalisation on a case-by-case basis in the following situations:
- Associated acute pathology (viral gastro-enteritis, bacterial infection, etc.)
- Age less than 3 months

In all other cases, the child may be treated at home, provided the parents are taught how to carry out treatment, and what signs of severity should lead to re-consultation.

Outpatient treatment

- Nasal irrigation with 0.9% NaCl before each feeding (demonstrate the technique to the mother)[a] .
- Small, frequent feedings to reduce vomiting triggered by bouts of coughing.
- Increased fluids if fever and/or significant secretions are present.
- Treat fever (Chapter 1).
- Handle the patient the patient as little as possible and avoid unnecessary procedures.

Hospitalisation

- In all cases:
  - Place the infant in a semi-reclining position (± 30°).
  - Nasal irrigation, small, frequent feeds, treatment of fever: as for outpatient treatment.
  - Gentle oro-pharyngeal suction if needed.
  - Monitor fluid intake: normal requirements are 80 to 100 ml/kg/day + 20 to 25 ml/kg/day with high fever or very profuse secretions.
- According to symptoms:
  - Humidified nasal oxygen if respiratory distress or $SpO_2 < 92\%$.
  - When there is vomiting or significant fatigue when sucking, fluid requirements may be administered by nasogastric tube (small volumes on a frequent basis) or the IV route, for the shortest possible time. Avoid breastfeeding or oral feeds in children with severe tachypnoea, but do not prolong NG feeds (respiratory compromise) or IV infusions any longer than necessary.
  - Bronchodilator therapy is not indicated but a trial treatment may be given in case of severe respiratory distress (**salbutamol** metered-dose inhaler, 100 micrograms/puff: 2 to 3 puffs with spacer, repeated 2 times at an interval of 30 minutes). If inhaled salbutamol appears effective in relieving symptoms, the treatment is continued (2 to 3 puffs every 6 hours in the acute phase, then gradual reduction as recovery takes place). If the trial is ineffective, the treatment is discontinued.
  - Antibiotics are not indicated unless there is concern about complications such as secondary bacterial pneumonia.

Prevention and control

The risk of transmission of the virus is increased in hospital settings:

- Children with bronchiolitis should be grouped together, away from other children (cohorting).
- As infection is most commonly transmitted by the hands, the most important prevention measure is hand-washing after any contact with patients, and objects or surfaces in contact with patients on which the virus may survive for several hours.
- In addition, staff should wear gowns, gloves and surgical masks when in contact with patients.

Footnotes

(a) Lie the child on his back, head turned to the side and instil 0.9% NaCl into the nose, one nostril at a time.

Acute pneumonia

- Pneumonia in children under 5 years of age
- Pneumonia in children over 5 years and adults
- Persistent pneumonia

Acute pneumonia is a viral, bacterial (pneumococcus, *Haemophilus influenzae*, staphylococcus, atypical bacteria) or parasitic (pneumocystosis) infection of the pulmonary alveoli.

Pneumonia in children under 5 years of age

The most common causes are viruses, pneumococcus and *Haemophilus influenzae*.

Clinical features

- Cough or difficulty breathing
- Fever often high (> 39 °C), but the child may present with low-grade fever or may have no fever (often a sign of serious illness)

Clinical examination must be done on a calm child in order to correctly count the respiratory rate and look for signs of severity.

- A child has tachypnoea (increased respiratory rate) if:
  RR ≥ 60 breaths/minute in children under 1 months
  RR ≥ 50 breaths/minute in children from 1 to 11 months
  RR ≥ 40 breaths/minute in children from 12 months to 5 years
- On pulmonary auscultation: dullness with diminished vesicular breath sounds, crepitations and sometimes bronchial breathing or normal pulmonary auscultation.
- Signs of severity (severe pneumonia):
  - Chest indrawing: the inferior thoracic wall depresses on inspiration as the superior abdomen expands
  - Cyanosis (lips, oral mucosa, fingernails) or SpO_2 < 90%
  - Nasal flaring
  - Altered consciousness (child is abnormally sleepy or difficult to wake)
  - Stridor (hoarse noise on inspiration)
  - Grunting (a short repetitive noise produced by a partial closure of the vocal cords) on expiration
  - Refusal to drink or feed
  - Children under 2 months
  - Severe malnutrition

Notes:
- In malnourished children, the RR thresholds should be decreased by 5 breaths/minute from those listed above.
- Chest indrawing is significant if it is clearly visible and present at all times. If it is observed when a child is upset or feeding and is not visible when the child is resting, there is no chest indrawing.
- In children under 2 months of age, moderate chest indrawing is normal as the thoracic wall is flexible.
- If only the soft tissues between the ribs or above the clavicles depress, there is no chest indrawing.

Consider also:
- Malaria in endemic areas, as it may also cause cough and tachypnoea.
- Staphylococcal pneumonia in patients with empyema or painful abdominal swelling and diarrhoea.
- Pneumocystosis in children with confirmed or suspected HIV infection (see HIV infection and AIDS, Chapter 8).
- Tuberculosis:
  - in a child with cough, fever and poor weight gain and a history of close contact with a tuberculous patient[a] . For the diagnosis, refer to the MSF handbook, Tuberculosis.
  - in the event of pneumonia complicated with empyema (pus in the pleural space).

Treatment

Severe pneumonia (inpatient treatment)

Children under 2 months

The first line treatment is the combination **ampicillin** slow IV (3 minutes) for 10 days + **gentamicin** slow IV (3 minutes) or IM for 5 days:

Children 0 - 7 days	< 2 kg	**ampicillin** 50 mg/kg every 12 hours + **gentamicin** 3 mg/kg once daily
	≥ 2 kg	**ampicillin** 50 mg/kg every 8 hours + **gentamicin** 5 mg/kg once daily
Children 8 days - < 1 month		**ampicillin** 50 mg/kg every 8 hours + **gentamicin** 5 mg/kg once daily
Children 1 month - < 2 months		**ampicillin** 50 mg/kg every 6 hours + **gentamicin** 6 mg/kg once daily

For ampicillin, IV route is preferred but IM route may be an alternative.

If ampicillin is not available, alternatives may be **cefotaxime** slow IV (3 minutes) or infusion (20 minutes) or IM for 10 days (for doses, see Meningitis, Chapter 7), or, as a last resort: **ceftriaxone** slow IV[b] (3 minutes) or infusion (30 minutes; 60 minutes in neonates) or IM: 50 mg/kg once daily for 10 days.

If the child's condition does not improve[c] after 48 hours of well administered treatment, add **cloxacillin** IV for 10 to 14 days:

Children 0 - 7 days	< 2 kg	**cloxacillin** 50 mg/kg every 12 hours
	≥ 2 kg	**cloxacillin** 50 mg/kg every 8 hours
Children > 7 days	< 2 kg	**cloxacillin** 50 mg/kg every 8 hours
	≥ 2 kg	**cloxacillin** 50 mg/kg every 6 hours

Children from 2 months to 5 years

The first line treatment is:
ceftriaxone IM or slow IV[b] (3 minutes): 50 mg/kg once daily
or
ampicillin slow IV (3 minutes) or IM: 50 mg/kg every 6 hours
+ **gentamicin** slow IV (3 minutes) or IM: 6 mg/kg once daily
Ampicillin is preferably administered in 4 divided doses. If the context does not permit it, the daily dose must be divided in at least 3 doses.

The treatment is administered by parenteral route for at least 3 days then, if the clinical condition has improved[c] and oral treatment can be tolerated, switch to **amoxicillin** PO: 30 mg/kg 3 times daily to complete 10 days of treatment.

If the child's condition deteriorates or does not improve after 48 hours of correct administration, add **cloxacillin** IV: 25 to 50 mg/kg every 6 hours. After clinical improvement and 3 days with no fever, switch to **amoxicillin/clavulanic acid** (**co-amoxiclav**) PO to complete 10 to 14 days of treatment. Use formulations in a ratio of 8:1 or 7:1 exclusively. The dose is expressed in amoxicillin: 50 mg/kg 2 times daily.

If the child's condition does not improve after 48 hours with ceftriaxone + cloxacillin, consider tuberculosis. For the diagnosis, refer to the guide Tuberculosis, MSF.

If tuberculosis is unlikely, continue with ceftriaxone + cloxacillin and add azithromycin (see Atypical pneumonia).

Notes:
- For malnourished children, refer to specific protocol.
- In the event of moderate-large empyema, assess if drainage is required. Administer antibiotics active against pneumococci and staphylococci (see Staphylococcal pneumonia).

Adjuvant therapy

- Fever: paracetamol PO (Chapter 1).
- Infants: keep warm.
- Install on an incline (head elevated) or in semi-sitting position.
- Clear the airway (nasal irrigation with 0.9% sodium chloride if needed).
- Oxygen at the flow rate required to maintain $SpO_2 \geq 90\%$ or, if pulse oxymeter is not available, minimum 1 litre/minute.
- Maintain adequate hydration and nutrition:
  - In children with severe respiratory difficulty: place an IV line and give 70% of normal maintenance fluids. Resume oral feeding as soon as possible (no severe respiratory difficulty, ability to eat normally).
    Use a nasogastric tube only if an IV line cannot be established: children under 12 months: 5 ml/kg/hour; children over 12 months: 3 to 4 ml/kg/hour; alternate milk and water. Resume normal oral feeding as soon as possible.
  - In the absence of severe respiratory difficulty: breastfeed on demand; milk/food and water by spoon on demand.
  - ORS when required (Dehydration, Chapter 1).

Pneumonia with no signs of severity

Children under 2 months

Admit the child for inpatient care and treat for severe pneumonia.

Children from 2 months to 5 years

Treat as outpatient, except infants.
amoxicillin PO: 30 mg/kg 3 times daily for 5 days

Follow-up in 48 to 72 hours or sooner if the child's condition deteriorates:
- If the condition is improving[c] : continue with the same antibiotic to complete treatment.
- If there is no improvement after 3 days of correct administration: add azithromycin (see Atypical pneumonia).
- If the condition is deteriorating: hospitalise and treat as severe pneumonia.

Footnotes

(a) Contact is defined as living in the same household, or in close and regular contact with any known or suspected tuberculous case within the last 12 months.

(b) The solvent of ceftriaxone for IM injection contains lidocaine. Ceftriaxone reconstituted using this solvent must never be administered by IV route. For IV administration, water for injection must always be used.

(c) Improvement criteria include: fever reduction, diminished respiratory distress, improved SpO_2, improved appetite and/or activity.

Pneumonia in children over 5 years and adults

The most common causes are viruses, pneumococcus, and *Mycoplasma pneumoniae*.

Clinical features

- Cough, with or without purulent sputum, fever, thoracic pain, tachypnoea
- On pulmonary auscultation: decreased vesicular breath sounds, dullness, localised foci of crepitations, sometimes bronchial wheeze.

Sudden onset with high fever (higher than 39 °C), thoracic pain and oral herpes are suggestive of pneumococcal infection. Symptoms may be confusing, particularly in children with abdominal pain, meningeal syndrome, etc.

Signs of severity (severe pneumonia) include:

- Cyanosis (lips, oral mucosa, fingernails)
- Nasal flaring
- Intercostal or subclavial indrawing
- RR > 30 breaths/minute
- Heart rate > 125 beats/minute
- Altered level of consciousness (drowsiness, confusion)

Patients at risk include the older patients, patients with heart failure, sickle cell disease or severe chronic bronchitis; immunocompromised patients (severe malnutrition, HIV infection with CD4 < 200).

Treatment

Severe pneumonia (inpatient treatment)

ceftriaxone IM or slow IV[a] (3 minutes)
Children: 50 mg/kg once daily
Adults: 1 g once daily
The treatment is given by parenteral route for at least 3 days then, if the clinical condition has improved[b] and oral treatment can be tolerated, switch to **amoxicillin** PO to complete 7 to 10 days of treatment:
Children: 30 mg/kg 3 times daily (max. 3 g daily)
Adults: 1 g 3 times daily
or
ampicillin slow IV (3 minutes) or IM
Children: 50 mg/kg every 6 hours
Adults: 1 g every 6 to 8 hours
Ampicillin is preferably administered in 4 divided doses. If the context does not permit it, the daily dose must be divided in at least 3 doses.
The treatment is given by parenteral route for at least 3 days then, if the clinical condition has improved[b] and oral treatment can be tolerated, switch to the oral route with amoxicillin PO as above, to complete 7 to 10 days of treatment.

If the clinical condition deteriorates or does not improve after 48 hours of correct administration, administer ceftriaxone as above + **cloxacillin** IV infusion:
Children: 25 to 50 mg/kg every 6 hours
Adults: 2 g every 6 hours
After clinical improvement and 3 days with no fever, switch to **amoxicillin/clavulanic acid** (**co-amoxiclav**) PO to complete 10 to 14 days of treatment. Use formulations in a ratio of 8:1 or 7:1 exclusively. The dose is expressed in amoxicillin:
Children < 40 kg: 50 mg/kg 2 times daily
Children ≥ 40 kg and adults:
Ratio 8:1: 3000 mg daily (2 tablets of 500/62.5 mg 3 times daily)
Ratio 7:1: 2625 mg daily (1 tablet of 875/125 mg 3 times daily)

If the clinical condition does not improve after 48 hours with ceftriaxone + cloxacillin, consider tuberculosis. For the diagnosis, refer to the guide Tuberculosis, MSF.

If tuberculosis is unlikely, continue with ceftriaxone + cloxacillin and add azithromycin (see Atypical pneumonia).

Adjuvant therapy

- Fever: paracetamol PO (Chapter 1).
- Clear the airway (nasal irrigation with 0.9% sodium chloride if needed).
- Oxygen at the flow rate required to maintain SpO_2 ≥ 90% or, if pulse oxymeter is not available, minimum 1 litre/minute.
- Maintain adequate hydration and nutrition.

Pneumonia without signs of severity (outpatient treatment)

amoxicillin PO
Children: 30 mg/kg 3 times daily (max. 3 g daily) for 5 days
Adults: 1 g 3 times daily for 5 days

Follow-up in 48 to 72 hours or sooner if the patient's condition deteriorates:
- If the condition is improving[b] : continue with the same antibiotic to complete treatment.
- If there is no improvement after 3 days of correct administration: add azithromycin (see Atypical pneumonia).
- If the condition is deteriorating: hospitalise and treat as severe pneumonia.

Footnotes

(a) The solvent of ceftriaxone for IM injection contains lidocaine. Ceftriaxone reconstituted using this solvent must never be administered by IV route. For IV administration, water for injection must always be used.

(b) Improvement criteria include: fever reduction, diminished respiratory distress, improved SpO_2, improved appetite and/or activity.

Persistent pneumonia

Last update: November 2022

In patients not responding to therapy, consider atypical pneumonia, tuberculosis, pneumocystosis (HIV infection and AIDS, Chapter 8).

Bacteria responsible for atypical pneumonia are mainly *Mycoplasma pneumoniae* and *Chlamydophila pneumoniae*. If suspected, one of the following antibiotics may be used:

First choice, **azithromycin** PO
Children: 10 mg/kg (max. 500 mg) once daily for 5 days
Adults: 500 mg on D1 then, 250 mg once daily from D2 to D5

If not available,
erythromycin PO
Children: 10 mg/kg (max. 500 mg) 4 times daily for 10 to 14 days
Adults: 500 mg 4 times daily for 10 to 14 days
or
doxycycline PO (except in pregnant or breastfeeding women)
Children under 45 kg: 2 to 2.2 mg/kg (max. 100 mg) 2 times daily for 10 to 14 days
Children 45 kg and over and adults: 100 mg 2 times daily for 10 to 14 days

Staphylococcal pneumonia

Pneumonia due to *Staphylococcus aureus* affecting young children, often those in a poor general condition (malnutrition, skin lesions, etc.). Staphylococcal pneumonia is a classic complication of measles.

Clinical features

- General signs: change in overall condition, pallor, high fever or hypothermia, frequently signs of shock; presence of skin lesions (point of bacterial entry), however, skin lesions may be absent.
- Gastrointestinal signs: nausea, vomiting, diarrhoea, painful abdominal distention.
- Respiratory signs: dry cough, tachypnoea, signs of distress (nasal flaring, chest indrawing). Pulmonary auscultation is often normal; sometimes dullness indicating pleural effusion.

Paraclinical investigations

- Chest x-ray (if available): may show multilobar consolidation, cavitation, pneumatoceles, spontaneous pneumothorax.

Treatment

Treatment is urgent as patients deteriorate quickly: hospitalise.

- Antibiotic treatment: if staphylococcal aetiology cannot be confirmed or while waiting for confirmation, a broad spectrum antibiotic therapy is recommended:
  ceftriaxone IM or slow IV[a] (at least 3 minutes): 50 mg/kg once daily
  + **cloxacillin** IV infusion (60 minutes)[b]
  Neonates 0 to 7 days (< 2 kg): 50 mg/kg every 12 hours
  Neonates 0 to 7 days (≥ 2 kg): 50 mg/kg every 8 hours
  Neonates 8 days to < 1 month (< 2 kg): 50 mg/kg every 8 hours
  Neonates 8 days to < 1 month (≥ 2 kg): 50 mg/kg every 6 hours
  Children 1 month and over: 25 to 50 mg/kg every 6 hours (max. 8 g daily)

  After clinical improvement[c] , 3 days with no fever, and drain removal if any, switch to **amoxicillin/clavulanic acid** PO to complete 10 to 14 days. Use formulations in a ratio of 8:1 or 7:1 exclusively. The dose is expressed in amoxicillin: 50 mg/kg 2 times daily

  In the event of large empyema: same treatment but switch to the oral route after 7 days with no fever and treat for 3 weeks.

  Clindamycin IV may be an alternative to cloxacillin: 10 mg/kg every 8 hours then switch to clindamycin PO at the same dose, according to the criteria above.

- Fever: paracetamol (Chapter 1).

- Hydration by oral route or infusion or nasogastric tube depending on clinical condition.

- Oxygen at the flow rate required to maintain SpO_2 ≥ 90% or, if pulse oxymeter is not available, minimum 1 litre/minute.

- Local disinfection of skin lesions.

- If there is significant pleural effusion: pleural tap with drainage (for pyopneumothorax; insert 2 drains, one anterior and one posterior) or without drainage (for suppurative pleurisy, make repetitive taps with an IV catheter).

Clinical evolution

- There is a serious risk of decompensation from pneumothorax or suppurative pleurisy or pyopneumothorax.
- On a paediatric ward, adequate equipment for urgent pleural drainage should always be available.

Footnotes

(a) The solvent of ceftriaxone for IM injection contains lidocaine. Ceftriaxone reconstituted using this solvent must never be administered by IV route. For IV administration, water for injection must always be used.

(b) Cloxacillin powder for injection should be reconstituted in 4 ml of water for injection. Then dilute each dose of cloxacillin in 5 ml/kg of 0.9% sodium chloride or 5 % glucose in children less than 20 kg and in a bag of 100 ml of 0.9% sodium chloride or 5% glucose in children 20 kg and over and in adults.

(c) Improvement criteria include: fever reduction, diminished respiratory distress, improved SpO2, improved appetite and/or activity.

Asthma

Last updated: June 2023

- Acute asthma (asthma attack)
- Chronic asthma

Asthma is a chronic inflammatory disorder of the airways associated with airway hyperresponsiveness that leads to recurrent episodes of wheezing, breathlessness, chest tightness and coughing. These episodes are usually associated with airflow obstruction within the lung, often reversible, either spontaneously or with treatment.

Factors that precipitate/aggravate asthma include: allergens, infection, exercise, drugs (aspirin), tobacco, etc. Symptoms are sometimes worse at night.

In children up to 5 years, most initial episodes of asthma-like symptoms are associated with a respiratory tract infection, with no symptoms between infections. Wheezing episodes usually become less frequent with time; most of these children do not develop asthma.

Acute asthma (asthma attack)

Last updated: June 2023

Asthma attack is a substantial worsening of asthma symptoms. The severity and duration of attacks are variable and unpredictable.

Assessment of the severity of asthma attack

The severity of the asthma attack must be rapidly evaluated by the following clinical criteria. Not all signs are necessarily present.

Assessment of severity in children over 2 years and adults[1][2][3]

Mild or moderate attack	Severe or life-threatening attack
Able to talk in sentences	**Cannot complete sentences in one breath** or **Too breathless to talk or feed**
Mild or moderate increase of **respiratory rate (RR)**	Very high **RR** Children 2-5 years: > 40/minute Children > 5 years and adults: > 30/minute
Normal or mild increase of **heart rate (HR)** Children 2-3 years: ≤ 180/minute Children 4-5 years: ≤ 150/minute Children > 5 years and adults: ≤ 120/minute	Very high **HR** Children 2-3 years: > 180/minute Children 4-5 years: > 150/minute Children > 5 years and adults: > 120/minute
SpO_2 ≥ 90% (≥ 92% for children 2-5 years)	**SpO_2** < 90% (< 92% for children 2-5 years)
and **No criteria of severe or life-threatening attack**	**Signs of life-threatening attack**: Altered level of consciousness (drowsiness, confusion, coma) Exhaustion Silent chest Cyanosis Arrhythmia or hypotension in adults

Treatment

Reassure the patient. Treatment and follow-up depend on the severity of the attack and the patient's response:

Mild to moderate attack

- Place the patient in a 1/2 sitting position.
- Administer:

- **salbutamol** metered-dose inhaler (MDI) 100 micrograms/puff: 2 to 10 puffs every 20 minutes during the first hour. In children, use a spacer[a] (use face mask in children under 3 years). Single puffs should be given one at a time, let the child breathe 4 to 5 times from the spacer before repeating the procedure. A spacer can also be used in adults to increase effectiveness.
- **prednisolone** PO: one dose of 1 to 2 mg/kg (max. 50 mg) for children over 5 years and adults
- **oxygen** if $SpO_2 < 94\%$[b].

- If the attack is completely resolved:
  - Observe the patient for 1 hour (4 hours if they live far from the health centre) then give outpatient treatment: **salbutamol** MDI for 24 to 48 hours (2 to 4 puffs every 4 to 6 hours depending on clinical evolution) and **prednisolone** PO (same dose as above once daily) to complete 5 days of treatment.
  - Reassess after 1 to 2 days: address any identified risk factor, reassess need for salbutamol and long-term treatment. If the patient is already receiving long-term treatment, reevaluate the severity of the asthma (see Chronic asthma), review compliance and correct use of medications and adjust treatment if necessary.
- If the attack is only partially resolved, continue with **salbutamol** MDI (2 to 10 puffs every 1 to 4 hours) until symptoms subside. For children up to 5 years, administer one dose of **prednisolone** PO as above if symptoms recur within 3 to 4 hours. When the attack is completely resolved, proceed as above.
- If symptoms worsen or do not improve, treat as severe attack.

Severe attack

- Hospitalise[c]; place the patient in a 1/2 sitting position.
- Administer:
  - **oxygen** to maintain SpO_2 between 94 and 98%[b].
  - **salbutamol** + **ipratropium** nebuliser solutions using a nebuliser (continue oxygen via nasal cannula during nebulisation):

Children < 5 years	**salbutamol** 2.5 mg (1.25 ml) + **ipratropium** 0.25 mg (1 ml) every 20 minutes for the first hour
Children 5 to 11 years	**salbutamol** 2.5 to 5 mg (1.25 to 2.5 ml) + **ipratropium** 0.5 mg (2 ml) every 20 minutes for the first hour
Children 12 years and over and adults	**salbutamol** 5 mg (2.5 ml) + **ipratropium** 0.5 mg (2 ml) every 20 minutes for the first hour

    The two solutions should be mixed in the drug reservoir of the nebuliser. Assess symptoms at the end of each nebulisation.
    If there is no nebuliser, use **salbutamol** MDI (same dose as for mild to moderate attack) and **ipratropium** MDI 20 micrograms/puff, 4 to 8 puffs every 20 minutes for the first hour.
  - **prednisolone** PO: one dose of 1 to 2 mg/kg (max. 50 mg)
    If prednisolone is not available, or if the patient cannot take oral treatment, administer:
    - Children: **dexamethasone** PO/IV/IM, one dose of 0.15 to 0.6 mg/kg (max. 16 mg)
    - Adults: **hydrocortisone** IV, 4 mg/kg (max. 100 mg) every 6 hours for 24 hours
- If symptoms do not improve after one hour:
  - transfer to intensive care unit
  - insert an IV line
  - **oxygen** to maintain SpO_2 between 94 and 98%[b]
  - continue **salbutamol** (solution for nebuliser) without ipratropium, and corticosteroids as above.
  - administer one dose of **magnesium sulfate** by IV infusion in 0.9% sodium chloride over 20 minutes, monitoring blood pressure:

    - Children: 40 mg/kg (max. 2 g)
    - Adults: 2 g
- If symptoms improve: continue **salbutamol** (solution for nebuliser) every 1 to 4 hours (depending on symptoms) and **oxygen** as above. Assess symptoms at the end of each nebulisation. When possible, switch to **salbutamol** MDI and continue as for mild to moderate attack.
- If the attack is completely resolved, observe the patient for at least 4 hours. Continue the treatment with **salbutamol** (MDI) and **prednisolone** PO and reassess as for a mild to moderate attack.

Notes:
- In pregnant women, treatment is the same as for adults. In mild or moderate asthma attacks, administering oxygen reduces the risk of foetal hypoxia.
- For all patients, irrespective of the severity of the asthma attack, look for underlying lung infection and treat accordingly.

Footnotes

(a) If a conventional spacer is not available, use a 500 ml plastic bottle: insert the mouthpiece of the inhaler into a hole made in the bottom of the bottle (the seal should be as tight as possible). The patient breathes from the mouth of the bottle in the same way as they would with a spacer. The use of a plastic cup instead of a spacer is not recommended (ineffective).

(b) If pulse oxymetry is not available, administer oxygen continuously in case of moderate, severe or life-threatening attack.

(c) If signs of life-threatening attack, transfer to intensive care unit as soon as possible.

References

1. British guideline on the management of asthma. A national clinical guideline First published 2003. Revised edition published July 2019. https://www.sign.ac.uk/our-guidelines/british-guideline-on-the-management-of-asthma/ [Accessed 12 January 2023]

2. Global INitiative for Asthma. Global Strategy for Asthma Management and Prevention. 2022 update. https://ginasthma.org/gina-reports/ [Accessed 12 January 2023]

3. WHO Pocket book of primary health care for children and adolescents: guidelines for health promotion, disease prevention and management from the newborn period to adolescence. WHO Regional Office for Europe; 2022. https://www.who.int/europe/publications/i/item/9789289057622 [Accessed 12 January 2023]

Chronic asthma

Last updated: June 2023

Clinical features

- Asthma should be suspected in patients with recurrent respiratory symptoms (wheezing, chest tightness, shortness of breath and/or cough) of variable frequency, severity and duration, disturbing sleep, and causing the patient to sit up to breathe. These symptoms may appear during or after exercise.
- Chest auscultation may be normal or demonstrate diffuse sibilant wheezes.
- A personal or family history of atopy (eczema, allergic rhinitis/conjunctivitis) or a family history of asthma increases probability of asthma but their absence does not exclude asthma.
- Patients with typical symptoms of asthma and a history of disease that is characteristic of asthma should be considered as having asthma after exclusion of other diagnoses.
- Any identified asthma risk factor (e.g. allergen, pollution, tobacco smoke exposure) should be eliminated where possible. The assessment of the frequency of symptoms and limitations of daily activities determines the treatment.

Treatment

The mainstay of long-term treatment are inhaled corticosteroids (ICS) and long-acting beta-2 agonists (LABA). LABAs should never be used alone but always in combination with an ICS. Combination inhalers are preferred, when available. In addition to long-term treatment, salbutamol (short-acting beta-2 agonist, SABA) and combination inhalers can be used to reduce bronchoconstriction if the patient is symptomatic.
Treatment is started at the step most appropriate to initial severity then, re-evaluated and adjusted according to clinical response. An intervening severe asthma attack or loss of control necessitates treatment reassessment.
The inhaler is chosen according to age. In children, a spacer should be used. Instructions on inhaler technique and information on asthma attack symptoms should be provided.

Long-term treatment of asthma according to severity in children 6 years and over and adults[1][2]

Symptoms	Children 6 to 11 years	Children ≥ 12 years and adults
Intermittent asthma • Daytime symptoms < 2 times monthly • Normal daily activities	**salbutamol** when symptomatic	**beclometasone/formoterol** when symptomatic
		OR **beclometasone** + **salbutamol** when symptomatic[a]
Mild persistent asthma • Daytime symptoms ≥ 2 times monthly • Symptoms may affect daily activities	**beclometasone** (low dose) daily AND **salbutamol** when symptomatic	**beclometasone/formoterol** when symptomatic
		OR **beclometasone** (low dose) daily AND **salbutamol** when symptomatic
Moderate persistent asthma • Daytime symptoms most days OR nighttime symptoms ≥ once weekly • Symptoms affect daily activities	**beclometasone** (low dose) + **salmeterol** daily[b] AND **salbutamol** when symptomatic	**beclometasone/formoterol** (low dose) daily AND **beclometasone/formoterol** when symptomatic
	OR **budesonide/formoterol** (very low dose) daily AND **budesonide/formoterol** when symptomatic	OR **beclometasone** (low dose) + **salmeterol** daily[b] AND **salbutamol** when symptomatic
Severe persistent asthma • Daily daytime symptoms OR very frequent nighttime symptoms • Daily activities very limited by symptoms	**beclometasone** (medium dose) + **salmeterol** daily AND **salbutamol** when symptomatic	**beclometasone/formoterol** (medium dose) daily AND **beclometasone/formoterol** when symptomatic
	OR **budesonide/formoterol** (low dose) daily AND **budesonide/formoterol** when symptomatic	OR **beclometasone** (medium dose) + **salmeterol** daily[c] AND **salbutamol** when symptomatic

(a) Salbutamol should be taken just before beclometasone, or together if a combination inhaler is available.

(b) If salmeterol is not available, use beclometasone medium-dose.

(c) If salmeterol is not available, use beclometasone high-dose.

The doses vary according to the severity of asthma. Find the lowest possible effective dose necessary to both relieve symptoms and avoid local and systemic adverse effects.

beclometasone MDI (ICS):

	Children 6 to 11 years	**Children ≥ 12 years and adults**
When symptomatic	–	200 to 500 micrograms
Long-term treatment		
Low dose	50 to 100 micrograms 2 times daily	100 to 250 micrograms 2 times daily
Medium dose	150 to 200 micrograms 2 times daily	300 to 500 micrograms 2 times daily
High dose	–	> 500 micrograms 2 times daily

In all cases, do not exceed 2000 micrograms daily.

The number of puffs of beclometasone depends on its concentration in the inhaled aerosol: 50, 100 or 250 micrograms per puff.

salbutamol MDI 100 micrograms/puff (SABA):
- Children and adults: 2 to 4 puffs up to 4 times daily if necessary

salmeterol MDI 25 micrograms/puff (LABA):
- Children 6 to 11 years: 2 puffs 2 times daily (max. 4 puffs daily)
- Children 12 years and over and adults: 2 to 4 puffs 2 times daily (max. 8 puffs daily)

budesonide/formoterol MDI 80/4.5 micrograms/puff (ICS/LABA combination):
- Children 6 to 11 years:
  - when symptomatic: 1 puff
  - long-term treatment, very low-dose: 1 puff once daily
  - long-term treatment, low-dose: 1 puff 2 times daily

In all cases, do not exceed 8 puffs daily.

beclometasone/formoterol MDI 100/6 micrograms/puff (ICS/LABA combination):
- Children 12 years and over and adults:
  - when symptomatic: 1 puff
  - long-term treatment, low-dose: 1 puff 2 times daily
  - long-term treatment, medium-dose: 2 puffs 2 times daily

In all cases, do not exceed 8 puffs daily.

Do not restrict exercise. If exercise is a trigger for asthma attacks, administer 1 or 2 puffs of salbutamol or beclometasone/formoterol 10 minutes beforehand.

In pregnant women, poorly controlled asthma increases the risk of pre-eclampsia, eclampsia, haemorrhage, in utero growth retardation, premature delivery, neonatal hypoxia and perinatal mortality. Long-term treatment should be continued under close monitoring.

If symptoms have not been well controlled for a period of 2 to 3 months, check inhalation technique and adherence before changing to a stronger treatment.

If symptoms have been well controlled for a period of at least 3 months (the patient is asymptomatic or the asthma attacks are well controlled): try a step-wise reduction in medication.

References

1. Global INitiative for Asthma. Global Strategy for Asthma Management and Prevention. 2022 update. https://ginasthma.org/gina-reports/ [Accessed 23 January 2023]

2. WHO Pocket book of primary health care for children and adolescents: guidelines for health promotion, disease prevention and management from the newborn period to adolescence. WHO Regional Office for Europe; 2022. https://www.who.int/europe/publications/i/item/9789289057622 [Accessed 23 January 2023]

Pulmonary tuberculosis

Pulmonary tuberculosis is a bacterial infection due to *Mycobacterium tuberculosis*, spread from person to person through inhalation of infected respiratory droplets.
After infection, *M. tuberculosis* multiplies slowly in the lungs and is usually eliminated spontaneously or lies dormant. Only 10% of cases develop active tuberculosis. The risk of progressing to active tuberculosis is higher in immunocompromised patients. In certain countries, half of newly diagnosed tuberculosis patients are co-infected with HIV[1] .

For more information on tuberculosis, refer to the guide Tuberculosis, MSF.

Clinical features

- Prolonged cough (> 2 weeks) with or without sputum production and/or haemoptysis, prolonged fever, night sweats, anorexia, weight loss, chest pain and fatigue.
- Differential diagnosis includes pneumonia, chronic obstructive pulmonary disease (COPD), lung cancer, pulmonary distomatosis (Flukes, Chapter 6) and melioidosis (Southeast Asia).

In an endemic area, the diagnosis of tuberculosis is to be considered, in any patient consulting for respiratory symptoms for over 2 weeks who does not respond to non-specific antibacterial treatment.

Laboratory

- In the general population: Xpert® MTB/RIF test which simultaneously detects *M. tuberculosis* (MTB) in sputum and resistance to rifampicin (RIF). If not available perform sputum smear microscopy[2] .
- If HIV co-infection suspected or diagnosed: Xpert® MTB/RIF test and point-of-care, urine LF-LAM (lateral flow urine lipoarabinomannan assay)[2] .

Treatment

For pulmonary tuberculosis, the standard treatment is a combination of four antituberculosis drugs (isoniazid, rifampicin, pyrazinamide, ethambutol). The regimen is organised into 2 phases (initial phase and continuation phase) and lasts 6 months.

If the strain is drug-resistant, the treatment is longer and different drug combinations are used.

It takes significant investment to cure tuberculosis, both from the patient and the medical team. Only uninterrupted treatment will lead to cure and prevent the development of resistance. It is essential that the patient understands the importance of treatment adherence and has access to correct case management until treatment is completed.

Prevention

- BCG vaccination in neonates: provides 59% protection against pulmonary tuberculosis[3] .
- Infection control in healthcare settings: standard precautions and airborne precautions for confirmed or suspected cases.
- Close contacts: isoniazid preventive therapy for 6 months.

References

1. World Health Organization. Global tuberculosis report 2018. https://apps.who.int/iris/handle/10665/274453 [Accessed 21 October 2019]

2. Global Laboratory Initiative. GLI model TB diagnostic algorithms. 2018. http://www.stoptb.org/wg/gli/assets/documents/GLI_algorithms.pdf [Accessed 21 October 2019]

3. World Health Organization. Weekly epidemiological record/Relevé épidémiologique hebdomadaire 23rd February 2018, 93rd year/23 Février 2018, 93e année. No 8, 2018, 93, 73–96. https://www.who.int/immunization/policy/position_papers/bcg/en/ [Accessed 21 October 2019]

Chapter 3: Gastrointestinal disorders

Acute diarrhoea

Shigellosis

Amoebiasis

Disorders of the stomach and duodenum

- Gastro-oesophageal reflux
- Gastric and duodenal ulcers in adults
- Dyspepsia

Stomatitis

- Oral and oropharyngeal candidiasis
- Oral herpes
- Other infectious causes
- Stomatitis from scurvy (vitamin C deficiency)
- Other lesions resulting from a nutritional deficiency